

The Lower Critical Endpoint of Exponential-Feedback Transseries

Uniform coefficient asymptotics, deletion of the largest action,
Landau cutoff profiles, and the compound-Poisson boundary

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Abstract

We study the lower endpoint of the critical power-law feedback equation $U(q) = A_\varepsilon(qe^{U(q)})$, where $A_\varepsilon(z) = (\text{Li}_{2+\varepsilon}(z) + P(z))/(\zeta(1+\varepsilon) + P'(1))$, $\varepsilon > 0$, and P changes a fixed finite prefix while preserving nonnegative coefficients. This is the $\alpha \downarrow 1$ half of Question 4 in the repository article *Beyond Finite-Action Folds*. Writing $\alpha = 1 + \varepsilon$, $\beta = 1/\alpha$, $K_\varepsilon = \Gamma(-\alpha)/(\zeta(\alpha) + P'(1))$, $B = (nK_\varepsilon)^\beta$, and $\rho_\varepsilon = e^{-A_\varepsilon(1)}$, we prove, uniformly down to arbitrarily small positive ε ,

$$[q^n]U(q) \sim \frac{\rho_\varepsilon^{-n}}{nB} \frac{\beta}{\Gamma(1-\beta)}.$$

A branch-cut integral retains the vanishing factor ε in both the main term and its error; it yields an explicit first correction and an arbitrary fixed-order expansion. The critical inverse has a convergent parameter-uniform Hahn chart, whose renormalized endpoint is a dilogarithm with its logarithmic singularity.

For the associated Poisson action process, conditioned on total weight n , deleting the largest action leaves the unconditioned independent Poisson process in total variation, for every joint limit $n \rightarrow \infty$, $\varepsilon \downarrow 0$. When $n\varepsilon \rightarrow \infty$, the largest action satisfies

$$\frac{J-B}{\varepsilon B} + \log \varepsilon \implies -Z, \quad \mathbb{E}e^{itZ} = \exp\{-\frac{\pi}{2}|t| - it \log |t|\}.$$

This gives a sharp Landau-distribution cutoff window and its quantiles. When $n\varepsilon \rightarrow \lambda < \infty$, the deficit $n - J$ instead converges in total variation to a discrete compound-Poisson law. These limits match as $\lambda \rightarrow \infty$. All proofs are conventional; reproducible exact finite checks and numerical diagnostics accompany the paper.

Status. This is a model-specific research manuscript with proofs, not a claim of a general transseries breakthrough or established publication priority. The classical analytic and condensation mechanisms are credited. No new Lean verification is claimed.

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1 The repository question and the contribution

The critical countable-action model in ProveIt’s article *Beyond Finite-Action Folds: Critical Hahn Transseries, Stable Sector Asymptotics, and Sharp Action Budgets* has a power-law exponent $1 < \alpha < 2$. Its Question 4 asks for joint limits at both endpoints. The editorial note at that question explicitly records that the later endpoint papers treat $\alpha \uparrow 2$, but not $\alpha \downarrow 1$ [1]. The present paper addresses the lower endpoint for the exact power-law tail with an arbitrary fixed admissible finite prefix. The repository snapshot used throughout is

3d5973524506411392a911470b5ddc35521568ea.

The index and the question were read directly at that revision [2, 1]. This is a statement about that snapshot, not a claim that no other unpublished work exists.

There is a genuine obstruction to substituting $\alpha = 1$ into a fixed-exponent answer. The first moment of $j^{-1-\alpha}$ diverges as $\alpha \downarrow 1$. Critical normalization therefore sends each fixed action weight to zero. At the same time the central stable-density constant tends to zero. A local limit theorem with an absolute error $o(1)$ is not enough to divide by that constant, and a fixed- α cutoff limit need not identify the new transition window.

The principal contributions are the following linked statements. First, a *relative*, parameter-uniform coefficient theorem includes arbitrarily small $\alpha - 1$, even values much smaller than any exponential in n . Second, a total-variation deletion theorem describes the *entire* remaining action process, not only the maximum. Third, the sharp continuous cutoff window becomes a reflected, completely asymmetric stable law of index one; a separate discrete compound-Poisson boundary is necessary when $n(\alpha - 1)$ is bounded. A convergent critical chart, an all-order coefficient algorithm, and a precise matching of the two probability limits complete the picture.

1.1 Classical background and the novelty boundary

The substitution $w = qe^U$ and Lagrange inversion are classical. The relation of such equations to random allocations and conditioned branching structures is treated extensively by Janson [5]. Hankel contours and singularity transfer are established analytic tools [4]; the exact polylogarithm expansion used here is the standard continuation formula [3]. The principle that a rare large sum is carried by one exceptional summand, with the other summands asymptotically unaffected, is a classical condensation mechanism; Armendáriz and Loulakis prove general conditional distribution results of that kind [6]. We do not claim these mechanisms as new discoveries.

What is proved here is their explicit, jointly varying realization at a *critically normalized, degenerating mean endpoint*. The proofs retain the constants and scales that disappear in fixed-parameter statements. In particular, no fixed-distribution big-jump theorem or fixed- α local limit theorem is invoked to justify a triangular limit outside its hypotheses. The analytical coefficient estimate and the deletion density are established directly below.

1.2 Scope of the answer

The tail in this paper is exactly $j^{-2-\varepsilon}$ after a fixed prefix. The critical coupling is exact. We allow the prefix coefficients to have either sign as perturbations, but the resulting action weights must be nonnegative. We do not cover arbitrary slowly varying tails, a prefix whose length grows with n , noncritical coupling windows, or unrestricted Borel summability and resurgence. Those are separate questions. Nor is there a nonzero finite-mean critical model obtained simply by setting $\varepsilon = 0$ before renormalizing.

Editorial note (ProveIt, 2026-09-29). The same question is posed as Question 10 of `Confluent_Critical_Transseries_Exponent_Two_Boundary/` (a compensated core, and whether conditioning couples the largest actions to the total, for the coupling $1/\zeta(\alpha)$, the case $P = 0$ here) and as Question 8 of `Logarithmic_Critical_Endpoint_Lambert_Charts/` (renormalized coupling, critical coordinate and action budget). On the critical path treated here they are answered by the Landau limit of the cloud sum on the scale εB (Theorem 8.3), by the deletion of exactly one largest action (Theorem 7.2), and by the coupling nc_ε , the chart coordinate T and the cutoff budget of Section 8. The interior-fold side $c > 1/\zeta(1 + \varepsilon)$ is treated in `Lower_Critical_Endpoint_Landau_Transseries/`; the subcritical side stays open.

2 Model and exact identities

Throughout, $\mathbb{N} = \{0, 1, 2, \dots\}$; empty maxima and missing second maxima are zero. Fix a real polynomial

$$P(z) = \sum_{j=1}^{J_*} p_j z^j, \quad p_k^{(\cdot)} := \sum_{j=1}^{J_*} p_j j^k, \quad \kappa = P'(1) = p_1^{(\cdot)}. \quad (1)$$

The superscript in $p_k^{(\cdot)}$ distinguishes the finite prefix moment from the polynomial coefficient p_k . Assume that, for all sufficiently small $\varepsilon \geq 0$,

$$b_{j,\varepsilon} = j^{-2-\varepsilon} + p_j \geq 0 \quad (1 \leq j \leq J_*), \quad b_{1,\varepsilon} > 0, \quad (2)$$

where $p_j = 0$ for $j > J_*$. Thus the tail is positive and aperiodic. Put

$$\alpha = 1 + \varepsilon, \quad \beta = \frac{1}{\alpha}, \quad c_\varepsilon = \frac{1}{\zeta(1 + \varepsilon) + \kappa}, \quad (3)$$

$$A_\varepsilon(z) = c_\varepsilon (\text{Li}_{2+\varepsilon}(z) + P(z)) = \sum_{j \geq 1} a_{j,\varepsilon} z^j, \quad a_{j,\varepsilon} = c_\varepsilon b_{j,\varepsilon}, \quad (4)$$

$$d_\varepsilon = A_\varepsilon(1), \quad \rho_\varepsilon = e^{-d_\varepsilon}, \quad K_\varepsilon = c_\varepsilon \Gamma(-\alpha). \quad (5)$$

All these quantities are real, and $c_\varepsilon, K_\varepsilon > 0$ for small positive ε . Criticality is the exact identity

$$A'_\varepsilon(1) = \sum_{j \geq 1} j a_{j,\varepsilon} = 1. \quad (6)$$

The feedback solution is the unique formal series with zero constant term satisfying

$$U_\varepsilon(q) = A_\varepsilon(qe^{U_\varepsilon(q)}), \quad u_{n,\varepsilon} = [q^n]U_\varepsilon(q). \quad (7)$$

The analytic implicit function theorem also gives convergence near zero. The formal recursion has nonnegative coefficients, with $u_{n,\varepsilon} > 0$. On the positive branch, $q = we^{-A_\varepsilon(w)}$ is strictly increasing for $0 < w < 1$, since $wA'_\varepsilon(w) < A'_\varepsilon(1) = 1$. It tends to ρ_ε as $w \uparrow 1$, which fixes the real branch used in the local critical chart below.

Proposition 2.1 (Coefficient and conditioning identities). *For every $n \geq 1$,*

$$u_{n,\varepsilon} = \frac{1}{n} [z^n] e^{nA_\varepsilon(z)}. \quad (8)$$

Let $\mathcal{N}_n = (N_j)_{j \geq 1}$ be independent Poisson variables of means $\lambda_{j,n} = na_{j,\varepsilon}$, and set

$$S_n = \sum_{j \geq 1} j N_j, \quad M(\mathcal{N}_n) = \max\{j : N_j > 0\}, \quad (9)$$

with maximum zero for the empty process. Then

$$u_{n,\varepsilon} = \frac{\rho_\varepsilon^{-n}}{n} \mathbb{P}(S_n = n), \quad \mathbb{E}S_n = n. \quad (10)$$

If $u_{n,\varepsilon}^{(m)}$ is the coefficient obtained by retaining only actions $j \leq m$, then

$$\frac{u_{n,\varepsilon}^{(m)}}{u_{n,\varepsilon}} = \mathbb{P}(M(\mathcal{N}_n) \leq m \mid S_n = n). \quad (11)$$

Proof. Set $w = qe^{U_\varepsilon(q)}$. Then $w = qe^{A_\varepsilon(w)}$ and $U_\varepsilon = A_\varepsilon(w)$. Lagrange inversion yields

$$[q^n]A_\varepsilon(w(q)) = \frac{1}{n}[z^{n-1}]A'_\varepsilon(z)e^{nA_\varepsilon(z)} = \frac{1}{n}[z^n]e^{nA_\varepsilon(z)},$$

where the last step differentiates the exponential. The total Poisson intensity is $nd_\varepsilon < \infty$, so the process has finitely many atoms almost surely. Its probability generating function is $\exp(n(A_\varepsilon(z) - d_\varepsilon))$, proving (10). The mean is (6). For (11), independence of the omitted coordinates contributes $\exp(-n \sum_{j>m} a_{j,\varepsilon})$; multiplication by the truncated generating function leaves exactly $e^{-nd_\varepsilon} \exp(n \sum_{j \leq m} a_{j,\varepsilon} z^j)$. $\square$

An action is a part of size j counted by N_j . We use *condensation* below in the precise sense that, under $S_n = n$, a single exceptional action is separated from a much smaller residual maximum. It need not carry a positive fraction of n in every regime.

3 A convergent chart through the lower endpoint

Define the regular expansion coefficients

$$D_k(\varepsilon) = \zeta(2 + \varepsilon - k) + p_k^{(\cdot)} \quad (k \geq 2). \quad (12)$$

They are not moments of the infinite action distribution. For example, the actual second moment diverges, whereas $D_2(\varepsilon)$ is finite and may be negative. The continuation formula for the polylogarithm gives, for $|x| < 2\pi$ and $|\arg x| < \pi$,

$$A_\varepsilon(e^{-x}) = d_\varepsilon - x + K_\varepsilon x^\alpha + c_\varepsilon \sum_{k \geq 2} \frac{D_k(\varepsilon)(-x)^k}{k!}. \quad (13)$$

This is an exact convergent local expansion, not merely an asymptotic one [3].

Lemma 3.1 (Removable endpoint parameters). *The functions c_ε and K_ε extend holomorphically through $\varepsilon = 0$, with*

$$c_0 = 0, \quad c_\varepsilon = \varepsilon + O(\varepsilon^2), \quad K_0 = 1, \quad K_\varepsilon = 1 - (1 + \kappa)\varepsilon + O(\varepsilon^2). \quad (14)$$

On a fixed disk $|x| < r_0 < 2\pi$, the function

$$H_\varepsilon(x) = c_\varepsilon \sum_{k \geq 2} \frac{D_k(\varepsilon)(-1)^k x^{k-2}}{k!} \quad (15)$$

is jointly holomorphic in (ε, x) near $\varepsilon = 0$, and $H_\varepsilon(x) = O(\varepsilon)$ uniformly on smaller disks.

Proof. Use $\zeta(1 + \varepsilon) = \varepsilon^{-1} + \gamma + O(\varepsilon)$ and

$$\Gamma(-1 - \varepsilon) = \frac{\Gamma(1 - \varepsilon)}{\varepsilon(1 + \varepsilon)}, \quad \Gamma(1 - \varepsilon) = 1 + \gamma\varepsilon + O(\varepsilon^2).$$

The poles cancel in K_ε . The only zeta pole in the full polylogarithm expansion near this parameter is the linear term, which has already been removed by criticality. The remaining series converges locally uniformly; the polynomial prefix is entire after composition with e^{-x} . This proves the last assertions. $\square$

Write $w = e^{-x}$ and $t = \log(\rho_\varepsilon/q)$, with branches continued from positive $q < \rho_\varepsilon$. The exact normal form is

$$t = K_\varepsilon x^\alpha + x^2 H_\varepsilon(x), \quad U_\varepsilon = d_\varepsilon + t - x. \quad (16)$$

Theorem 3.2 (Uniform convergent Hahn chart). *There are neighborhoods of zero, independent of sufficiently small $\varepsilon \geq 0$, and a jointly holomorphic function $V(\varepsilon, T, R)$ such that*

$$T = (t/K_\varepsilon)^\beta, \quad R = T^{1-\varepsilon}, \quad (17)$$

$$x = TV(\varepsilon, T, T^{1-\varepsilon}), \quad V^\alpha + \frac{R}{K_\varepsilon} V^2 H_\varepsilon(TV) = 1. \quad (18)$$

Here $V(0, T, R) = 1$, and $V - 1$ is divisible by εR in the local holomorphic ring. In particular,

$$x = T - \frac{c_\varepsilon D_2(\varepsilon)}{2\alpha K_\varepsilon} T^{2-\varepsilon} + O(\varepsilon |T|^{3-\varepsilon} + \varepsilon^2 |T|^{3-2\varepsilon}) \quad (19)$$

on every sufficiently small fixed closed subsector. The full expansion is convergent and has support contained in

$$\left\{ \frac{1+k+(1-\varepsilon)m}{1+\varepsilon} : k, m \in \mathbb{N} \right\}. \quad (20)$$

Taylor truncations in the independent variables (T, R) have a geometric remainder on smaller polydisks, uniformly in ε .

Proof. Near $V = 1$ define V^α by the local logarithm. The left side of (18) minus one is jointly holomorphic and its derivative in V at $(T, R, V) = (0, 0, 1)$ is $1 + \varepsilon$, bounded away from zero. The parameter-dependent implicit function theorem applies on a uniform polydisk. At $\varepsilon = 0$, $H_0 = 0$ and the equation has solution $V = 1$. At $R = 0$ the same is true for every parameter. The Taylor expansion thus has the factor εR . Its coefficient linear in R is $-H_\varepsilon(T)/(\alpha K_\varepsilon)$ at $V = 1$; expanding $H_\varepsilon(T)$ at zero gives (19). Substitution of $R = T^{1-\varepsilon}$ gives (20). Cauchy estimates on a slightly larger polydisk give the geometric remainder. No divergent formal expansion is being summed in this statement. $\square$

The confluent coordinate can be written without hiding its large parameter combination:

$$T = t \exp \left(-\frac{\varepsilon}{1+\varepsilon} \log t - \frac{\log K_\varepsilon}{1+\varepsilon} \right). \quad (21)$$

Expanding the exponential in powers of ε is inappropriate when $\varepsilon \log(1/|t|)$ is not small. Keeping it exact is the lower-endpoint counterpart of a uniform critical chart.

Proposition 3.3 (Renormalized dilogarithmic endpoint). *On every compact subset of $|z| < 1$,*

$$\frac{U_\varepsilon(\rho_\varepsilon z)}{\varepsilon} \longrightarrow H_0^*(z) := \text{Li}_2(z) + P(z). \quad (22)$$

Its continuation near $z = e^{-t} = 1$ has expansion

$$H_0^*(e^{-t}) = \zeta(2) + P(1) + t \log t - (1 + \kappa)t + \sum_{k \geq 2} \frac{D_k(0)(-t)^k}{k!}. \quad (23)$$

Thus the nontrivial endpoint is logarithmic after division by ε ; the unscaled limit is zero.

Proof. On compact subdisks, $A_\varepsilon(z)/\varepsilon \rightarrow H_0^*(z)$ locally uniformly, while the implicit solution is $O(\varepsilon)$. Substituting into the feedback equation proves (22); $\rho_\varepsilon \rightarrow 1$ does not change it. To obtain (23), combine the cancellation of the linear zeta pole with the expansion of $\Gamma(-1-\varepsilon)t^{1+\varepsilon}$, or differentiate (16) at $\varepsilon = 0$. The polynomial adds $P(1) - \kappa t + \sum_{k \geq 2} p_k^{(\cdot)}(-t)^k/k!$. $\square$

4 The uniform coefficient theorem

For $n \geq 1$ define

$$B = B_{n,\varepsilon} = (nK_\varepsilon)^{1/(1+\varepsilon)}, \quad h = \frac{n}{B^2}, \quad J_r(\varepsilon) = \frac{\Gamma((r+1)/\alpha)}{\pi\alpha} \sin\left(\frac{\pi\varepsilon(r+1)}{\alpha}\right). \quad (24)$$

The basic constant has several equivalent forms:

$$J_0(\varepsilon) = \frac{\beta}{\Gamma(1-\beta)} = -\frac{1}{\Gamma(-\beta)} > 0, \quad J_0(\varepsilon) \sim \varepsilon. \quad (25)$$

For each fixed nonnegative integer r ,

$$J_r(\varepsilon) \sim (r+1)!\varepsilon, \quad \frac{J_r(\varepsilon)}{J_0(\varepsilon)} \rightarrow (r+1)!. \quad (26)$$

Individual J_r with large r can have either sign away from the endpoint; the formulas below retain their exact values.

Theorem 4.1 (Relative asymptotics uniform at a vanishing amplitude). *For each fixed admissible prefix P , there are constants $\varepsilon_* > 0$, $C > 0$, $\eta > 0$, and n_0 , with $\varepsilon_* < 1/4$, such that for all $n \geq n_0$ and $0 < \varepsilon \leq \varepsilon_*$,*

$$u_{n,\varepsilon} = \frac{\rho_\varepsilon^{-n} J_0(\varepsilon)}{nB} \left[1 + O\left(\varepsilon \frac{n}{B^2} + n^2 e^{-\eta n}\right) \right]. \quad (27)$$

The O -constant is independent of n and ε in this range. Consequently the leading equivalent holds for every joint limit $n \rightarrow \infty$, $\varepsilon \downarrow 0$, without a lower bound on $n\varepsilon$ or on ε itself.

The theorem is stronger than replacing a fixed-exponent stable law by its limit. The main probability is of order ε/B , not $1/B$. The proof in Section 5 keeps the small sine factor in the integral and also in its error. This is the key uniformity mechanism.

Editorial note (ProveIt, 2026-09-29). The leading term $\rho_\varepsilon^{-n} J_0(\varepsilon)/(nB)$ is the critical coefficient law `eq:critical-leading` of [1] (its Theorem `thm:phases`, stated for fixed $1 < \alpha < 2$) with $a = 1$, $\beta_c = c_\varepsilon$ and $B_{\beta_c} = K_\varepsilon$; Theorem 4.1 proves it uniformly as $\alpha \downarrow 1$. An independent package of the same batch, `Lower_Critical_Endpoint_Cauchy_Cutoff_Condensation/`, treats the same family at the same pinned revision and proves the same coefficient law for every rate of approach, with the same leading term but without an error rate. Its scales $d_n = (nc_\varepsilon/\varepsilon)^{1/(1+\varepsilon)}$ and $b_n = (nc_\varepsilon)^{1/(1+\varepsilon)}$ satisfy $B/d_n \rightarrow 1$ and $\varepsilon B/b_n \rightarrow 1$, and its truncated-mean centre satisfies $D_n = B + \varepsilon B \log(1/\varepsilon) + (1-\gamma)b_n + o(b_n)$. Its stable variable is $Z + 1 - \gamma$ for the standard Landau variable Z used here, so the two cutoff laws coincide. It also proves the

joint convergence of the residual extremes with the cloud, which Theorem 8.6 does not claim. Corollary 6.2 reproduces its recorded exact-recurrence coefficient ratios to three or four digits. Where the two articles overlap they state one theorem with two independent proofs, and every shared constant agrees.

Corollary 4.2 (Macroscopic crossover of sector coefficients). *Suppose $\varepsilon \rightarrow 0$ and $\varepsilon \log n \rightarrow \tau \in [0, \infty)$. Then*

$$\frac{B}{n} \rightarrow e^{-\tau}, \quad u_{n,\varepsilon} \sim \frac{\varepsilon e^\tau}{n^2} \rho_\varepsilon^{-n}. \quad (28)$$

When $\varepsilon \log n \rightarrow \infty$, the general equivalent (27) still applies, while $B/n \rightarrow 0$.

Proof. Take logarithms: $\log(B/n) = \beta \log K_\varepsilon - \varepsilon \beta \log n$. Use $K_\varepsilon \rightarrow 1$, $\beta \rightarrow 1$, and $J_0/\varepsilon \rightarrow 1$. $\square$

Corollary 4.3 (Microscopic coefficient boundary). *If $n\varepsilon \rightarrow \lambda \in [0, \infty)$, then*

$$u_{n,\varepsilon} \sim \frac{\varepsilon}{n^2} \exp\{\lambda(\zeta(2) + P(1))\}. \quad (29)$$

Proof. Now $\varepsilon \log n \rightarrow 0$, while $nd_\varepsilon = n\varepsilon(\zeta(2) + P(1)) + o(1)$. $\square$

5 Proof by a branch-cut integral

This section proves the uniform theorem without using a parameter-dependent local limit theorem. Let

$$R_\varepsilon(r) = c_\varepsilon \sum_{k \geq 2} \frac{D_k(\varepsilon) r^k}{k!}. \quad (30)$$

For fixed sufficiently small $r_0 > 0$, uniformly in the parameter,

$$|R_\varepsilon(r)| \leq C\varepsilon r^2 \quad (0 \leq r \leq r_0). \quad (31)$$

On the upper bank of the cut $z = e^r > 1$ one has $-\log z = -r - i0$. Equation (13) therefore gives

$$A_\varepsilon(e^r + i0) - d_\varepsilon - r = -K_\varepsilon r^\alpha \cos(\pi\varepsilon) + iK_\varepsilon r^\alpha \sin(\pi\varepsilon) + R_\varepsilon(r). \quad (32)$$

The regular part is the same real number on both banks.

Lemma 5.1 (Uniform cut representation). *After decreasing r_0 and ε_* if necessary,*

$$\mathbb{P}(S_n = n) = \frac{1}{\pi} \int_0^{r_0} e^{-nK_\varepsilon r^\alpha \cos(\pi\varepsilon) + nR_\varepsilon(r)} \sin(nK_\varepsilon r^\alpha \sin(\pi\varepsilon)) dr + E_{n,\varepsilon}, \quad (33)$$

$$|E_{n,\varepsilon}| \leq Cn\varepsilon e^{-\eta n}. \quad (34)$$

Proof. Start with the coefficient integral for $e^{n(A_\varepsilon(z) - d_\varepsilon)}$. The polylogarithm continues to the plane slit along $[1, \infty)$. Deform the circle about zero to the circle $|z| = e^{r_0}$ together with the two banks of the cut from 1 to e^{r_0} . A small circle about 1 contributes zero in the limit because the integrand is bounded there. Conjugacy of the banks and $dz/z = dr$ give (33) with the stated sign.

On the outer circle, including its two limiting endpoints on the cut, $\text{Li}_{2+\varepsilon}(z) + P(z)$ is uniformly bounded for small ε . Since $c_\varepsilon = O(\varepsilon)$, decrease ε_* until the real part of $A_\varepsilon(z) - d_\varepsilon - \log z$ is at most $-\eta < 0$ there. To preserve the small factor in the error, subtract the constant e^{-nd_ε} inside the coefficient integrand. Its integral around the full outer circle is zero for $n \geq 1$, and it has no cut jump. The inequality $|e^v - 1| \leq |v|e^{|v|}$ then bounds the remaining outer integral by $Cn\varepsilon e^{-\eta n}$. This subtraction is important when ε is much smaller than e^{-n} . $\square$

Lemma 5.2 (The elementary Hankel moments). *For $r \geq 0$ an integer and small positive ε ,*

$$\frac{1}{\pi} \int_0^\infty s^r e^{-s^\alpha \cos(\pi\varepsilon)} \sin(s^\alpha \sin(\pi\varepsilon)) ds = J_r(\varepsilon). \quad (35)$$

Moreover, for fixed r , the integral of the absolute value is $O_r(\varepsilon)$ uniformly in ε .

Proof. Put $y = s^\alpha$ and take the imaginary part of

$$\frac{1}{\pi\alpha} \int_0^\infty y^{(r+1)/\alpha-1} \exp(-e^{-i\pi\varepsilon} y) dy = \frac{\Gamma((r+1)/\alpha)}{\pi\alpha} e^{i\pi\varepsilon(r+1)/\alpha}.$$

The absolute bound follows from $|\sin(s^\alpha \sin \pi\varepsilon)| \leq C\varepsilon s^\alpha$ and a fixed positive lower bound for $\cos \pi\varepsilon$. Euler's reflection formula then gives (25), and Taylor expansion of the sine gives (26). $\square$

Proof of Theorem 4.1. In (33) set $s = Br$, so $nK_\varepsilon r^\alpha = s^\alpha$. The part with $R_\varepsilon = 0$, extended to infinity, is $J_0(\varepsilon)/B$. The extension error is $O(\varepsilon n e^{-\eta n})$, after adjusting constants, by the last bound in Lemma 5.2 and $(Br_0)^\alpha = nK_\varepsilon r_0^\alpha$.

Choose r_0 so small that (31) implies

$$|nR_\varepsilon(s/B)| \leq \frac{1}{2} \cos(\pi\varepsilon) s^\alpha \quad (0 \leq s \leq Br_0).$$

This follows since $r^2/r^\alpha = r^{2-\alpha}$ is small uniformly on that interval. Also $|nR_\varepsilon(s/B)| \leq C\varepsilon(n/B^2)s^2$. Hence

$$\begin{aligned} \frac{1}{B} \int_0^{Br_0} e^{-s^\alpha \cos(\pi\varepsilon)} |e^{nR_\varepsilon(s/B)} - 1| |\sin(s^\alpha \sin \pi\varepsilon)| ds \\ \leq \frac{C\varepsilon^2}{B} \frac{n}{B^2} \int_0^\infty e^{-cs^\alpha} s^{\alpha+2} ds \leq \frac{C\varepsilon^2}{B} \frac{n}{B^2}. \end{aligned}$$

The integral is bounded uniformly for $1 \leq \alpha \leq 1 + \varepsilon_*$. Divide by $J_0(\varepsilon)/B \asymp \varepsilon/B$. Since $B \leq Cn$, the outer error divided by the same main term is $O(n^2 e^{-\eta n})$. This proves (27) using (10). Finally, $h = n/B^2 \leq Cn^{-(1-\varepsilon_*)/(1+\varepsilon_*)} \rightarrow 0$ uniformly, so the stated joint-limit conclusion follows. $\square$

Remark 5.3 (What the proof avoids). The contour is a coefficient contour for the explicitly known function $\exp(nA_\varepsilon)$, not a conjectured global continuation of the inverse U_ε . The proof therefore does not require a uniform global inverse-function domain. It also does not divide an absolute stable approximation by a density constant tending to zero.

6 All fixed orders and the first correction

For a finite multi-index $\nu = (\nu_2, \nu_3, \dots)$ set

$$L(\nu) = \sum_{k \geq 2} \nu_k, \quad r(\nu) = \sum_{k \geq 2} k\nu_k, \quad d(\nu) = r(\nu) - L(\nu) = \sum_{k \geq 2} (k-1)\nu_k. \quad (36)$$

There are only finitely many multi-indices of a fixed cost d .

Theorem 6.1 (Uniform finite-order expansion). *For every fixed integer $N \geq 0$, uniformly in the range of Theorem 4.1,*

$$\begin{aligned} \frac{nB\rho_\varepsilon^n u_{n,\varepsilon}}{J_0(\varepsilon)} = \sum_{d(\nu) \leq N} \frac{J_{r(\nu)}(\varepsilon)}{J_0(\varepsilon)} \frac{(nc_\varepsilon)^{L(\nu)}}{B^{r(\nu)}} \prod_{k \geq 2} \frac{D_k(\varepsilon)^{\nu_k}}{(k!)^{\nu_k} \nu_k!} \\ + O_N(\varepsilon h^{N+1} + n^2 e^{-\eta n}). \end{aligned} \quad (37)$$

The zero multi-index contributes one. The expansion is asymptotic to arbitrary fixed order; convergence as $N \rightarrow \infty$ is not asserted.

Proof. Expand $e^{nR_\varepsilon(s/B)}$ in (33). Its monomial belonging to ν has coefficient

$$\frac{(nc_\varepsilon)^{L(\nu)}}{B^{r(\nu)}} \prod_{k \geq 2} \frac{D_k(\varepsilon)^{\nu_k}}{(k!)^{\nu_k} \nu_k!} s^{r(\nu)}.$$

Lemma 5.2 integrates it exactly. To justify a fixed truncation, use the Taylor remainder for the analytic regular part on a smaller disk and the integral remainder for the exponential. Each nonconstant term carries at least one factor $c_\varepsilon = O(\varepsilon)$. Since $B \leq Cn$, a monomial of cost d satisfies

$$\frac{n^L}{B^r} = h^{r-L} \left(\frac{B}{n} \right)^{r-2L} \leq C_N h^d \quad (d \leq N+1).$$

More explicitly, for $N \geq 1$, first retain the terms $k \leq N+1$ in R_ε . Its omitted tail in the exponent has bound $C_N \varepsilon n B^{-N-2} s^{N+2}$, which is at most $C_N \varepsilon h^{N+1} s^{N+2}$. The retained exponent $Q(s)$ obeys $|Q(s)| \leq C_N \varepsilon h s^2$ on the cut. Expanding $e^{Q(s)}$ through power N leaves an error bounded by $C_N \varepsilon^{N+1} h^{N+1} s^{2N+2} e^{|Q(s)|}$. Among those finitely many powers, the monomials of cost greater than N have the same required h^{N+1} factor. By making the cut radius smaller if needed, both $e^{|Q(s)|}$ and the analogous factor for the analytic tail are absorbed by the decaying e^{-cs^α} , just as in the preceding proof. The case $N = 0$ is Theorem 4.1. The sine gives another factor $O(\varepsilon)$ before division by $J_0 \asymp \varepsilon$. The relative remainder is therefore $O_N(\varepsilon h^{N+1})$. Polynomial factors in the exponentially small contour tails are absorbed by decreasing η . This proves (37). $\square$

Corollary 6.2 (An explicit first correction). *Let*

$$\Delta_{n,\varepsilon} = \frac{nc_\varepsilon D_2(\varepsilon)}{2B^2} \frac{J_2(\varepsilon)}{J_0(\varepsilon)}. \quad (38)$$

Then

$$u_{n,\varepsilon} = \frac{\rho_\varepsilon^{-n} J_0}{nB} \left[1 + \Delta_{n,\varepsilon} + O(\varepsilon h^2 + n^2 e^{-\eta n}) \right]. \quad (39)$$

As $\varepsilon \downarrow 0$,

$$\frac{\Delta_{n,\varepsilon}}{\varepsilon n / B^2} \longrightarrow 3 \left(-\frac{1}{2} + p_2^{(\cdot)} \right). \quad (40)$$

In particular, the unperturbed prefix gives $-3/2$.

Proof. At cost one the only multi-index is $\nu_2 = 1$. Apply $c_\varepsilon/\varepsilon \rightarrow 1$, $D_2(0) = -1/2 + p_2^{(\cdot)}$, and $J_2/J_0 \rightarrow 6$. $\square$

For clarity, the next cost adds exactly

$$\frac{nc_\varepsilon D_3}{6B^3} \frac{J_3}{J_0} + \frac{n^2 c_\varepsilon^2 D_2^2}{8B^4} \frac{J_4}{J_0}. \quad (41)$$

This is a finite, directly implementable expansion. The first prefix moment enters the exact core K_ε ; the second enters the first relative correction. There is no unexplained claim of prefix independence at orders where a prefix moment is visible.

7 Deleting the condensate: the entire cloud in total variation

Let $\mathcal{N}_n^*$ have the law of $\mathcal{N}_n$ conditioned on $S_n = n$. Remove one occurrence of its largest action and denote the remaining multiset by $\mathcal{R}_n$. The definition is unambiguous even when the maximum is repeated: remove one atom at that size. Write J_n for the removed size. All probability statements about J_n henceforth use the conditioned law.

Lemma 7.1 (A deterministic first-order cloud deficit). *For every joint limit $n \rightarrow \infty$, $\varepsilon \downarrow 0$, under the unconditioned law,*

$$\frac{S_n - n}{B} \Rightarrow -1, \quad \frac{M(\mathcal{N}_n)}{B} \rightarrow 0 \quad \text{in probability.} \quad (42)$$

Proof. The logarithm of the characteristic function of the centered sum is, using (13),

$$\log \mathbb{E} e^{it(S_n - n)/B} = (-it)^\alpha + O_t\left(\frac{nc_\varepsilon}{B^2}\right). \quad (43)$$

The error tends to zero uniformly, and $(-it)^\alpha \rightarrow -it$, with the principal logarithm and continuous value at $t = 0$. Lévy's continuity theorem gives the first assertion. For each fixed $d > 0$, once $dB > J_*$,

$$\mathbb{P}(M(\mathcal{N}_n) > dB) \leq nc_\varepsilon \sum_{j > dB} j^{-1-\alpha} \leq C_d nc_\varepsilon B^{-\alpha} = C_d \frac{c_\varepsilon}{K_\varepsilon} = O_d(\varepsilon).$$

This proves the second assertion. $\square$

Theorem 7.2 (Total-variation deletion theorem). *For every joint limit $n \rightarrow \infty$, $\varepsilon \downarrow 0$,*

$$d_{\text{TV}}(\mathcal{L}(\mathcal{R}_n), \mathcal{L}(\mathcal{N}_n)) \rightarrow 0. \quad (44)$$

The conditioned largest action is unique with probability tending to one, and

$$\frac{J_n}{B} \rightarrow 1, \quad \frac{J_n^{(2)}}{B} \rightarrow 0 \quad \text{in probability,} \quad (45)$$

where $J_n^{(2)}$ is the second largest action, with multiplicities.

Proof. Let $p_n = \mathbb{P}(S_n = n)$. For an unconditioned multiset ν with total weight s and maximum m , define

$$W_n(\nu) = \frac{\lambda_{n-s,n}}{p_n} \mathbf{1}_{\{n-s > m, n-s \geq 1\}}, \quad (46)$$

where a rate with nonpositive index is zero. The Poisson deletion identity gives, for every bounded nonnegative function f ,

$$\mathbb{E}[f(\mathcal{R}_n); \mathcal{N}_n^* \text{ has a unique maximum}] = \mathbb{E}[f(\mathcal{N}_n)W_n(\mathcal{N}_n)]. \quad (47)$$

Indeed, adding a strictly larger atom of size j to ν multiplies its Poisson mass by $\lambda_{j,n}$; the total-weight condition forces $j = n - s$. Alternatively, sum the usual Poisson identity $\mathbb{E}[N_j f(\mathcal{N}_n - \delta_j)] = \lambda_{j,n} \mathbb{E}[f(\mathcal{N}_n)]$ with the strict-largest condition. In particular, $\mathbb{E}W_n \leq 1$.

By Lemma 7.1, $j = n - S_n$ satisfies $j/B \rightarrow 1$ in probability and exceeds the unconditioned maximum with probability tending to one. It is eventually beyond the fixed prefix, except on an event of probability tending to zero. There,

$$\frac{\lambda_{j,n}}{p_n} = \frac{c_\varepsilon/K_\varepsilon}{J_0(\varepsilon)} \left(\frac{B}{j}\right)^{1+\alpha} (1 + o(1)) \rightarrow 1, \quad (48)$$

because $c_\varepsilon/K_\varepsilon = 1/\Gamma(-1-\varepsilon) \sim \varepsilon$ and $J_0 \sim \varepsilon$. The $o(1)$ in the denominator is the relative error proved in Theorem 4.1. Hence $W_n \rightarrow 1$ in probability.

This convergence automatically upgrades to L^1 : bounded convergence in probability gives $\mathbb{E} \min(W_n, 1) \rightarrow 1$; together with $\mathbb{E} W_n \leq 1$ it yields $\mathbb{E} W_n \rightarrow 1$ and

$$\mathbb{E}|W_n - 1| = \mathbb{E} W_n + 1 - 2\mathbb{E} \min(W_n, 1) \rightarrow 0.$$

Equation (47) now proves both that repeated maxima have probability tending to zero and that the subprobability law on unique maxima converges in total variation to the unconditioned law. Adding the vanishing exceptional mass proves (44).

Under the conditioned law, $J_n = n - S(\mathcal{R}_n)$ and $J_n^{(2)} = M(\mathcal{R}_n)$. Total variation allows these n -dependent measurable maps, so (45) follows from Lemma 7.1. $\square$

Remark 7.3 (Why the mean is not contradicted). The unconditioned process has mean total weight exactly n , although its typical total weight is $n - B + o_{\mathbb{P}}(B)$. Rare very large jumps carry the missing mean. Total variation does not preserve expectations of unbounded, non-uniformly-integrable observables. Thus deletion convergence is compatible with the deterministic bound $S(\mathcal{R}_n) < n$.

The proof is a triangular-array version of a classical big-jump deletion mechanism. Its short final density argument is possible only after the vanishing central probability has been estimated *relatively*. This connects the analytic part of the paper to the probabilistic part without an unproved uniform conditioning step.

8 Landau fluctuations and sharp continuous budgets

Definition 8.1 (The endpoint stable law). Let Z be the probability law with characteristic function

$$\varphi_Z(t) = \exp\{-it \log(-it)\} = \exp\{-\frac{\pi}{2}|t| - it \log|t|\}, \quad \varphi_Z(0) = 1. \quad (49)$$

Let $G(z) = \mathbb{P}(-Z \leq z)$ be its reflected distribution function.

The logarithm in the first expression is principal. The specified characteristic function fixes location and scale; it avoids ambiguities among conventions for the Landau distribution. Existence also follows from the compound-Poisson approximation in Section 10. Since $|\varphi_Z(t)| = e^{-\pi|t|/2}$, Fourier inversion gives a smooth real-analytic density. It is nonnegative and not identically zero; therefore it cannot vanish on an interval. The distribution functions of Z and $-Z$ are continuous and strictly increasing.

Lemma 8.2 (The necessary microscopic scale). *If $\varepsilon \rightarrow 0$ and $n\varepsilon \rightarrow \infty$, then*

$$a_n := \varepsilon B \rightarrow \infty, \quad \frac{n}{\varepsilon B^2} \rightarrow 0. \quad (50)$$

Proof. The bounded factors K_ε^β can be suppressed. If $\varepsilon < n^{-1/2}$, then $\varepsilon \log n \rightarrow 0$, $B \sim n$, and the two assertions reduce to $n\varepsilon \rightarrow \infty$. If $\varepsilon \geq n^{-1/2}$, then $\varepsilon B \geq n^{1/2-o(1)}$ and $n/(\varepsilon B^2) \leq n^{-1/2+o(1)}$. These two cases cover all indices. $\square$

Theorem 8.3 (Landau law for the cloud and condensate). *Assume $n \rightarrow \infty$, $\varepsilon \downarrow 0$, and $n\varepsilon \rightarrow \infty$. Under the unconditioned law,*

$$Z_n := \frac{S_n - n + B}{\varepsilon B} - \log \varepsilon \implies Z. \quad (51)$$

Under the conditioned law,

$$\frac{J_n - B}{\varepsilon B} + \log \varepsilon \implies -Z. \quad (52)$$

The same limiting law applies for every fixed admissible prefix, provided the exact prefix-dependent K_ε is used in B .

Proof. Using (13) at frequency $t/(\varepsilon B)$ gives

$$\log \mathbb{E} e^{itZ_n} = \left(\frac{-it}{\varepsilon} \right)^{1+\varepsilon} + \frac{it}{\varepsilon} - it \log \varepsilon + O_t \left(\frac{n}{\varepsilon B^2} \right). \quad (53)$$

The remainder tends to zero by Lemma 8.2. For each fixed $t \neq 0$, write $L = \log(-it) - \log \varepsilon$. Then

$$\left(\frac{-it}{\varepsilon} \right)^{1+\varepsilon} = \frac{-it}{\varepsilon} e^{\varepsilon L} = \frac{-it}{\varepsilon} - itL + O_t(\varepsilon(1 + |\log \varepsilon|)^2).$$

Substitution into (53) leaves $-it \log(-it) + o(1)$. Continuity at zero and Lévy's theorem prove (51). For the conditional statement use $J_n = n - S(\mathcal{R}_n)$ and Theorem 7.2. The identity of the normalized variables has exactly the minus sign displayed in (52). $\square$

The logarithmic displacement is essential. The leading location is $B - \varepsilon B \log \varepsilon$, not simply B , on the fluctuation scale εB . Because $-\log \varepsilon \rightarrow \infty$, dropping it changes every fixed cutoff quantile.

Theorem 8.4 (Sharp cutoff profile and quantiles). *Under the hypotheses of Theorem 8.3, put*

$$m_n(z) = \lfloor B + \varepsilon B(z - \log \varepsilon) \rfloor. \quad (54)$$

For every real z ,

$$\frac{u_{n,\varepsilon}^{(m_n(z))}}{u_{n,\varepsilon}} \longrightarrow G(z). \quad (55)$$

For $\theta \in (0, 1)$ let $q_\theta = G^{-1}(\theta)$ and define the minimal action budget

$$m_{n,\varepsilon}^*(\theta) = \min\{m \in \mathbb{N} : u_{n,\varepsilon}^{(m)} / u_{n,\varepsilon} \geq \theta\}. \quad (56)$$

Then

$$m_{n,\varepsilon}^*(\theta) = B + \varepsilon B(q_\theta - \log \varepsilon) + o(\varepsilon B). \quad (57)$$

Proof. Equation (11), Theorem 8.3, and $\varepsilon B \rightarrow \infty$ give (55); the floor is negligible. Continuity and strict increase of G imply quantile convergence: for any fixed $d > 0$, the retained fraction at $m_n(q_\theta - d)$ is eventually less than θ , and at $m_n(q_\theta + d)$ it is eventually greater. Monotonicity sandwiches the minimal budget, and then $d \downarrow 0$. $\square$

Corollary 8.5 (Coarse cutoff step). *For fixed $0 < s < 1$ the retained fraction at $m = \lfloor sB \rfloor$ tends to zero, and for fixed $s > 1$ it tends to one. At $m = \lfloor B \rfloor$ it also tends to zero under $n\varepsilon \rightarrow \infty$.*

Proof. The strict inequalities follow from $J_n/B \rightarrow 1$. At $m = B$, the normalized threshold in (52) is $\log \varepsilon + o(1) \rightarrow -\infty$; tightness proves the final assertion. $\square$

8.1 The residual extremes

Theorem 8.6 (Poisson residual extremes). *Under $n\varepsilon \rightarrow \infty$, the point process of residual action sizes, rescaled by $a_n = \varepsilon B$, converges on $(0, \infty)$, vaguely on sets bounded away from zero, to a Poisson point process of intensity $x^{-2} dx$. In particular,*

$$\mathbb{P}\left(\frac{J_n^{(2)}}{\varepsilon B} \leq x\right) \longrightarrow e^{-1/x} \quad (x > 0). \quad (58)$$

No independence between this extreme process and the Landau cloud sum is claimed.

Proof. For the unconditioned process the rescaled intensity on a compact interval in $(0, \infty)$ is a Riemann sum with factor

$$nc_\varepsilon a_n^{-\alpha} = \frac{c_\varepsilon}{K_\varepsilon} \varepsilon^{-\alpha} \longrightarrow 1,$$

since $c_\varepsilon/K_\varepsilon \sim \varepsilon$ and $\varepsilon^{-\varepsilon} \rightarrow 1$. The lattice spacing $1/a_n$ tends to zero. Independence of disjoint Poisson counts gives convergence of Laplace functionals. The finite prefix disappears from every such interval. For the maximum, the exact omitted intensity satisfies $n \sum_{j>a_n x} a_{j,\varepsilon} \rightarrow 1/x$, so its zero-count probability tends to $e^{-1/x}$. Transfer both conclusions to the residual process using Theorem 7.2. $\square$

9 The discrete compound-Poisson boundary

Let $\mathcal{C}_\lambda$ be the independent Poisson action process with rates $\lambda b_{j,0} = \lambda(j^{-2} + p_j)$, and let C_λ be its total weight. Its total intensity is finite even though its mean is infinite when $\lambda > 0$. Its generating function is

$$\mathbb{E}z^{C_\lambda} = \exp\{\lambda(\text{Li}_2(z) + P(z) - \zeta(2) - P(1))\}. \quad (59)$$

For $\lambda = 0$ this is the point mass at zero.

Theorem 9.1 (Microscopic cloud and deficit in total variation). *If $n\varepsilon \rightarrow \lambda \in [0, \infty)$, then*

$$d_{\text{TV}}(\mathcal{L}(\mathcal{R}_n), \mathcal{L}(\mathcal{C}_\lambda)) \rightarrow 0, \quad d_{\text{TV}}(\mathcal{L}(n - J_n), \mathcal{L}(C_\lambda)) \rightarrow 0. \quad (60)$$

Consequently, for every fixed integer $k \geq 0$,

$$\frac{u_{n,\varepsilon}^{(n-k)}}{u_{n,\varepsilon}} \longrightarrow \mathbb{P}(C_\lambda \geq k). \quad (61)$$

Proof. For each j , $na_{j,\varepsilon} \rightarrow \lambda b_{j,0}$. Since nc_ε is bounded, dominated convergence with the summable tail j^{-2} gives

$$\sum_{j \geq 1} |na_{j,\varepsilon} - \lambda b_{j,0}| \longrightarrow 0.$$

Couple independent Poisson counts by their common intensities and independent excess counts. The probability of any disagreement is at most the sum above, so the unconditioned processes converge in total variation. Combine this with Theorem 7.2. The total-weight map preserves the total-variation bound, proving the deficit claim. Finally $J_n \leq n - k$ is exactly $n - J_n \geq k$. $\square$

For example, the limiting retained fraction when the single largest possible action is omitted is

$$\frac{u_{n,\varepsilon}^{(n-1)}}{u_{n,\varepsilon}} \longrightarrow 1 - e^{-\lambda(\zeta(2)+P(1))}. \quad (62)$$

When $n\varepsilon \rightarrow 0$, essentially all the coefficient comes from configurations whose largest action is exactly n ; every cutoff below n loses asymptotically all of it. This is not a continuous cutoff profile. For a prescribed retained fraction at a lattice boundary, quantile statements must respect atoms: one may compare the discrete tails in (61), but cannot assert convergence of the minimal integer budget at a threshold equal to a limiting tail probability without additional information.

10 Matching the two endpoint descriptions

Proposition 10.1 (Compound-Poisson to Landau matching). *As $\lambda \rightarrow \infty$,*

$$\frac{C_\lambda}{\lambda} - \log \lambda - (1 + \kappa) \implies Z. \quad (63)$$

Proof. Put $x = -it/\lambda$ in (23). The logarithm of the characteristic function of C_λ/λ is

$$\lambda(H_0^*(e^{it/\lambda}) - H_0^*(1)) = -it \log(-it) + it \log \lambda + it(1 + \kappa) + O_t(\lambda^{-1}).$$

Subtracting the two centering terms gives (49) in the limit. Lévy's theorem proves the claim and, independently, establishes existence of the probability law in Definition 8.1. $\square$

This calculation checks the otherwise delicate constant in the logarithmic displacement. For example, in the matching overlap

$$\varepsilon \log^2 n \longrightarrow 0, \quad n\varepsilon \longrightarrow \infty, \quad (64)$$

write $\lambda_n = n\varepsilon$. Expansion of the exact core gives

$$B = n - n\varepsilon(\log n + 1 + \kappa) + O(n\varepsilon^2(1 + \log n)^2). \quad (65)$$

It follows that the conditional deficit, on the scale λ_n , is centered at

$$n - B + \varepsilon B \log \varepsilon = \lambda_n(\log \lambda_n + 1 + \kappa) + o(\lambda_n). \quad (66)$$

Together with Theorem 8.3 this gives the same centered Landau law as (63). Condition (64) is stated explicitly: without it, replacing the exact B by its first-order expansion need not be accurate on the fluctuation scale. The exact-core formulation requires no such extra condition.

10.1 A regime map

The two combinations $\varepsilon \log n$ and $n\varepsilon$ play different roles. The first controls the fraction of total mass in the exceptional action; the second determines whether the residual cloud is continuous after rescaling or remains discrete. Table 1 summarizes this. All entries assume $\varepsilon \downarrow 0$ and $n \rightarrow \infty$.

The leading coefficient remains the single expression $\rho_\varepsilon^{-n} J_0 / (nB)$ throughout this table. What changes is the structure of conditioning and the resolution needed for the action budget. This is why a coefficient equivalent alone would not answer the repository's cutoff question.

Parameter regime	Largest action	Residual description and budget
$\varepsilon \log n \rightarrow \tau \in (0, \infty)$	$J_n/n \rightarrow e^{-\tau}$	Landau window of width εB , centered at $B - \varepsilon B \log \varepsilon$.
$\varepsilon \log n \rightarrow \infty$	$J_n \sim B = o(n)$	One action is still separated from the residual maximum by a factor of order $1/\varepsilon$; the same Landau window applies.
$\varepsilon \log n \rightarrow 0, n\varepsilon \rightarrow \infty$	$J_n/n \rightarrow 1$	A diverging residual cloud; retain the exact B for fluctuation-scale accuracy.
$n\varepsilon \rightarrow \lambda \in (0, \infty)$	$J_n = n - O_{\mathbb{P}}(1)$	Deficit tends to C_λ ; cutoff transitions are discrete.
$n\varepsilon \rightarrow 0$	$\mathbb{P}(J_n = n) \rightarrow 1$	Omitting action n loses asymptotically the whole coefficient.

Table 1: The lower-endpoint regimes. “ $J_n \sim B$ ” here means $J_n/B \rightarrow 1$ in probability, not convergence of expectations.

11 Reproducible checks and numerical evidence

The package contains `verify.py` and its recorded output `verification_results.json`. The program performs three distinct kinds of checks. First, 385 exact rational assertions compare a direct feedback recursion with Lagrange coefficients, enumerate finite action multisets, and check the Poisson deletion and cutoff identities. Second, an independent 80-digit recurrence at $n = 100$, $\varepsilon = 0.05$ agrees with the log-space numerical implementation to a relative difference of approximately 2.19×10^{-15} . Third, it evaluates the asymptotic and probability formulas at finite parameters. These last calculations are floating-point diagnostics, not proofs and not interval certificates.

11.1 Coefficient recurrence and normalization

For a fixed target index n , let $p_k = \mathbb{P}(S_n = k)$. Then

$$p_0 = e^{-nd_\varepsilon}, \quad p_k = \frac{n}{k} \sum_{j=1}^k j a_{j,\varepsilon} p_{k-j} \quad (1 \leq k \leq n). \quad (67)$$

The implementation uses log-sum-exp, so it does not require an extended floating-point exponent range. For a cutoff m , retain $j \leq m$ in the sum but *keep the same full* $p_0 = e^{-nd_\varepsilon}$. The resulting coefficient is $\mathbb{P}(S_n = k, N_j = 0 \text{ for } j > m)$, not the normalized probability under a different truncated process. This distinction is necessary for (11).

Table 2 compares the leading and corrected formulas for the unperturbed prefix. All values in the tables below are generated from the recorded JSON output; no qualitative convergence assertion is inferred from a few finite ratios.

The program also checks the prefix $P(z) = z/2 - z^2/16$. Here $\kappa = 3/8$, $p_2^{(\cdot)} = 1/4$, and the limiting first-correction constant is $-3/4$, rather than $-3/2$. This checks the explicit prefix sensitivity rather than suppressing it.

11.2 Landau profile and finite-parameter limitations

Fourier inversion gives a convenient formula for the distribution of Z :

$$F_Z(x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty e^{-\pi t/2} \frac{\sin(t(x + \log t))}{t} dt, \quad G(z) = 1 - F_Z(-z). \quad (68)$$

ε	n	Recurrence / leading	Recurrence / corrected	Observed correction / predicted
0.200	128	0.9923082678	1.0000206141	0.997348
0.200	2048	0.9987860446	1.0000006387	0.999475
0.050	128	0.9991009265	0.9999994150	1.000651
0.050	2048	0.9999268717	0.9999999976	1.000033
0.005	128	0.9999384519	0.9999998976	1.001666
0.005	2048	0.9999960518	0.9999999996	1.000106

Table 2: The first correction materially improves the coefficient approximation. “Exact recurrence” means the exact recurrence evaluated in floating point, not exact real arithmetic.

The integrable logarithmic singularity at zero is handled by splitting the integral. The exponential tail permits a controlled tail estimate, although the supplied quadrature itself does not use directed rounding. The numerical quantiles are

$$q_{0.1} \approx -11.6492846845, \quad q_{0.5} \approx -1.3557804210, \quad q_{0.9} \approx 1.0922545281. \quad (69)$$

Table 3 deliberately shows that a sharp limit theorem need not give a high-accuracy finite- ε profile. The errors are still visible at the displayed parameters. No rate for Theorem 8.4 is claimed.

n	ε	z	cutoff $m_n(z)$	retained fraction	$G(z)$
1024	0.08	-2	597	0.56178994	0.42447207
1024	0.08	0	689	0.80100720	0.71316712
4096	0.04	-2	3008	0.52738142	0.42447207
4096	0.04	0	3237	0.76751330	0.71316712
8192	0.02	-2	6993	0.49261488	0.42447207
8192	0.02	0	7262	0.74368211	0.71316712

Table 3: Finite-parameter cutoff diagnostics. The discrepancies, rather than only favorable ratios, are retained in the report.

11.3 The discrete boundary

For $k < n/2$, the event $J_n = n - k$ has a unique largest atom, so its probability is exactly

$$\mathbb{P}(n - J_n = k) = \frac{na_{n-k,\varepsilon}\mathbb{P}(S_n = k)}{\mathbb{P}(S_n = n)}. \quad (70)$$

This allows a check of the finite- λ cloud without Monte Carlo conditioning on a rare event. For $P = 0$, $\lambda = 1/2$, the mass at zero should tend to $e^{-\pi^2/12} \approx 0.4393464341$. The computed values are 0.4343650385 at $n = 512$ and 0.4377984669 at $n = 2048$. The recorded output includes both prefixes, $\lambda = 1/2, 2$, and errors for deficits 0 through 15.

11.4 What has and has not been verified

The 385 rational assertions verify finite algebraic identities only. The numerical recurrences check the stated formulas and normalizations at their input parameters. The asymptotic theorems, total-variation convergence, and parameter-uniform errors are supported by the proofs in this paper, not by those tests. No interval-certified action budget has been produced, and no Lean file is included. The supplied sources are sufficient to rebuild the article and rerun every reported computation.

12 A formalization and implementation route

The repository separates the transseries foundations from its research reports [2]. Nothing in the present package should be marked as a Lean theorem merely because related foundational files exist. A practical formalization can nevertheless be decomposed into small interfaces.

The finite algebra comes first: formal coefficient extraction, Lagrange inversion in the form (8), the exponential coefficient recurrence, and the finite multiset cutoff identity. The next layer can treat a countable summable sequence of Poisson rates and prove the exact deletion density (47). The upgrade from $W_n \rightarrow 1$ in probability and $\mathbb{E}W_n \leq 1$ to L^1 convergence is a short, reusable lemma independent of transseries.

The analytic layer needs the polylogarithm continuation formula with uniform parameter estimates on a compact slit contour, followed by the cut deformation, the elementary gamma integral, and the uniform exponential remainder bound. It is useful to state these as an abstract “regular part plus one branch term” coefficient theorem before specializing to the zeta coefficients. Characteristic-function convergence and the exact-core expansion then provide the probability limits. Finally, quantile convergence follows from a general monotone CDF lemma.

The code accompanying this paper is an implementation of finite recurrences and diagnostics. It is not a placeholder formal proof, and it does not replace any of those analytical obligations.

13 Further research questions

1. Quantitative deletion and Landau approximation. The total-variation theorem here is qualitative. Derive an explicit bound for the deletion distance and for the cutoff CDF error, uniformly in both $n\varepsilon$ and $\varepsilon \log n$. The characteristic-function proof suggests contributions of order $\varepsilon \log^2(1/\varepsilon)$ and $n/(\varepsilon B^2)$, but smoothing and the deletion density introduce additional terms. These suggestions are not error bounds proved in this paper. Such a result would explain and improve the visible finite-parameter discrepancies in Table 3.

2. An interval-certified minimal action budget. Combine the exact positive recurrence with rigorous logarithmic arithmetic, certified zeta and gamma values, and a proved finite- n CDF bound. The goal is a program that returns the smallest cutoff with a guaranteed retained fraction, or a narrow certified bracket. The asymptotic quantile formula alone is not such a program.

3. Slowly varying and nonexact tails. Replace the exact tail by $j^{-2-\varepsilon}L_\varepsilon(j)$. Determine verifiable uniform regularity conditions under which the deletion theorem and Landau window persist, and identify the new centering constants. Without an analytic continuation of the generating function, the current cut proof is not available; a genuinely uniform local big-jump estimate would be needed.

4. Growing or parameter-singular finite cores. The proof allows any fixed admissible prefix, but not a prefix whose length tends to infinity or whose moments diverge with ε . Find sharp criteria separating harmless prefix changes from a change in the limiting cloud. In particular, decide when the analytic regular part remains negligible on the scale εB .

5. Expanding noncritical coupling windows. For $U = \theta A_\varepsilon(qe^U)$, criticality is $\theta = 1$. Study the joint variables $1 - \theta$, ε , and n . A deterministic shift of the cloud competes with

the exceptional action and can move the cutoff center by its entire width. The present exact critical normalization does not settle that crossover.

6. Uniform matching beyond the first Landau order. Construct a single approximation that retains the discrete compound-Poisson law for bounded $n\varepsilon$, has a computable first Landau correction for diverging $n\varepsilon$, and uses the exact core for arbitrary $\varepsilon \log n$. This would refine both boundary descriptions without imposing (64).

7. Large-order behavior of the endpoint expansion. The convergent Hahn chart and the fixed-order sector expansion are not the same kind of series. Determine the growth of the coefficients in (37), an optimal truncation rule, and any additional exponentially small sectors. No Borel or resurgence conclusion follows from fixed-order uniformity alone.

8. The unnormalized exponent-one model. At exactly $\alpha = 1$, the unnormalized kernel $a \operatorname{Li}_2(z) + P(z)$ has infinite first moment, and the normalization used here collapses to zero. Analyze its interior critical point as the coupling tends to zero, and match that problem to the renormalized endpoint studied here. It is a different order of limits, not a missing substitution in (27).

Editorial note (ProveIt, 2026-09-29). For $P = 0$ this model is treated in `Lower_Critical_Endpoint_Landau_Transseries/`: an interior fold at distance $\delta = -\log(1 - e^{-1/c})$, a Landau coefficient profile at $\lambda = nc\delta$, and a chart in $e^{-1/c}$. Its matching to the renormalized endpoint studied here is not addressed there.

9. Other analytic branch kernels and multivariate actions. Abstract the sine-factor argument to kernels with a branch exponent approaching an integer and a simultaneously vanishing normalization. Determine which multivariate constraints admit an analogue of the one-atom deletion density, and when several exceptional actions rather than one are forced by geometry.

14 Conclusion

The lower critical endpoint can be analyzed without a formal substitution into a fixed-exponent theorem. The relevant coefficient probability vanishes like ε/B , and a branch-cut proof preserves that small factor uniformly. Once the relative estimate is available, a direct deletion density proves convergence of the entire conditioned remainder to the unconditioned Poisson cloud. Its fluctuations produce a Landau cutoff window when $n\varepsilon$ diverges and a discrete compound-Poisson deficit when it does not.

These results answer the lower-endpoint chart, coefficient, and cutoff questions for the exact power-law model with a fixed admissible prefix. They also identify precisely what remains: quantitative conditional errors, certified budgets, nonexact tails, and noncritical or growing-core extensions. The distinction between a proved model-specific continuation and a general transseries theorem is maintained throughout.

A Endpoint constants and sign checks

The expansions required to audit the centering are

$$c_\varepsilon = \varepsilon - (\gamma + \kappa)\varepsilon^2 + O(\varepsilon^3), \quad (71)$$

$$K_\varepsilon = 1 - (1 + \kappa)\varepsilon + O(\varepsilon^2), \quad (72)$$

$$J_0(\varepsilon) = \varepsilon + (\gamma - 2)\varepsilon^2 + O(\varepsilon^3), \quad (73)$$

$$K_\varepsilon^{-\beta} J_0(\varepsilon) = \varepsilon[1 + (\gamma - 1 + \kappa)\varepsilon + O(\varepsilon^2)]. \quad (74)$$

The sign of the stable variable is checked directly from

$$-it \log(-it) = -\frac{\pi}{2}|t| - it \log|t|.$$

For the largest action the variable is reflected because the exact identity is $J_n = n - S(\mathcal{R}_n)$. Its quantile center therefore is $B + \varepsilon B(q_\theta - \log \varepsilon)$, with a plus sign before q_θ . The residual total weight uses the opposite sign.

There is also a finite- n perturbative check for $P = 0$. At first order in ε , $u_{n,\varepsilon} = \varepsilon/n^2 + O_n(\varepsilon^2)$. At second order,

$$\frac{n^2 u_{n,\varepsilon}}{\varepsilon} = 1 + \varepsilon \left(n H_{n-1}^{(2)} + 2H_{n-1} - \gamma - \log n \right) + O_n(\varepsilon^2), \quad (75)$$

where $H_m = \sum_{j \leq m} 1/j$ and $H_m^{(2)} = \sum_{j \leq m} 1/j^2$. Indeed,

$$[z^n] \text{Li}_2(z)^2 = \frac{2H_{n-1}^{(2)}}{n^2} + \frac{4H_{n-1}}{n^3},$$

which follows by partial fractions in j and $n - j$. Expanding the harmonic sums gives the relative correction $-3\varepsilon/(2n)$ at the endpoint, agreeing with (40). The O_n notation here is explicitly for fixed n ; this perturbative check is not used as a uniform proof.

B Rebuilding and source provenance

The archive contains the article source and compiled PDF, generated LaTeX tables, the verification program, its recorded JSON results, `README.md`, `requirements.txt`, and a build script. The computational commands are

```
python verify.py --output verification_results.json
python make_tables.py
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The `-quick` verification option omits the larger cutoff calculations, and should write to a different output filename when the recorded full output is to be preserved. No network access is required by the verification program. The numerical dependencies are NumPy, SciPy, and mpmath; exact assertions use Python's standard-library `fractions` module.

Editorial note (ProveIt, 2026-09-29). As filed, `verify.py` writes `build/rerun_results.json` by default and refuses to replace the recorded `verification_results.json` unless `-overwrite-recorded` is given, so the first command above needs that option; `make_tables.py` and `verify.py` write LF line endings on every platform, and `build.sh` regenerates the tables only when asked (`-tables`).

The repository question and model were read from the pinned `Critical_Hahn_Transseries_Beyond_Finite_Action_Folds/article.tex` under `Analysis/Transseries/docs/series-and-transseries/`. The motivating question is Section “Further research questions”, Question 4; the exact coefficient and Poisson identities appear in its model section. The fixed-exponent chart motivates Section 3, but all uniform endpoint, deletion, and cutoff arguments used here are proved in this manuscript. No assumption that the unreviewed predecessor is correct is needed to validate these proofs.

References

- [1] ProveIt repository, maintained by V. Reshetnikov, *Beyond Finite-Action Folds: Critical Hahn Transseries, Stable Sector Asymptotics, and Sharp Action Budgets*, editorial revision of 29 September 2026. Repository revision 3d5973524506411392a911470b5ddc35521568ea, `Analysis/Transseries/docs/series-and-transseries/Critical_Hahn_Transseries_Beyond_Finite_Action_Folds/article.tex`. <https://github.com/VladimirReshetnikov/ProveIt>.
- [2] ProveIt transseries research index, maintained by V. Reshetnikov, same revision, `Analysis/Transseries/docs/series-and-transseries/README.md`. <https://github.com/VladimirReshetnikov/ProveIt>.
- [3] NIST Digital Library of Mathematical Functions, Section 25.12, especially Eq. 25.12.12 (polylogarithm continuation). <https://dlmf.nist.gov/25.12>. Accessed 29 September 2026.
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